



SMOTE-WENN: Solving class imbalance and small sample problems by oversampling and distance scaling

Hongjiao Guan^{1,2,3} · Yingtao Zhang³ · Min Xian⁴ · H. D. Cheng⁵ · Xianglong Tang³

Published online: 25 September 2020

© Springer Science+Business Media, LLC, part of Springer Nature 2020

Abstract

Many practical applications suffer from imbalanced data classification, in which case the minority class has degraded recognition rate. The primary causes are the sample scarcity of the minority class and the intrinsic complex distribution characteristics of imbalanced datasets. The imbalanced classification problem is more serious on small sample datasets. To solve the problems of small sample and class imbalance, a hybrid resampling method is proposed. The proposed method combines an oversampling approach (synthetic minority oversampling technique, SMOTE) and a novel data cleaning approach (weighted edited nearest neighbor rule, WENN). First, SMOTE generates synthetic minority class examples using linear interpolation. Then, WENN detects and deletes unsafe majority and minority class examples using weighted distance function and k-nearest neighbor (kNN) rule. The weighted distance function scales up a commonly used distance by considering local imbalance and spacial sparsity. Extensive experiments over synthetic and real datasets validate the superiority of the proposed SMOTE-WENN compared with three state-of-the-art resampling methods.

Keywords Imbalanced data classification · Small sample datasets · Oversampling · Data cleaning

1 Introduction

Imbalanced data classification is common in practical applications [1, 2], such as medical diagnosis, defect prediction, etc. It has always been a challenge in data mining and machine learning. For two-class datasets, imbalance means that the number of one class (called positive or minority class) is far less than that of the other class (called negative or majority class). Imbalanced datasets lead to

the performance deterioration of traditional classification methods. Especially, the recognition rate of the minority class decreases seriously. However, the minority class is of our interest from an application point of view. Furthermore, the misclassification cost of the minority class is usually higher than that of the majority class.

The imbalanced classification problem can be explained in two aspects. One is due to inappropriate optimization metrics in traditional learning algorithms. These algorithms assume the same misclassification costs for different classes, and their goal is to minimize the error rate on the training data. This results in these algorithms favoring the majority class. The other cause is the imbalanced data itself. Class imbalance is the most explicit obstacle for imbalanced classification. Imbalance ratio (IR) is used to measure the imbalance degree, which is defined as the sample number of the majority class divided by that of the minority class. Research has shown that the larger IR, the worse performance [3, 4]. Class imbalance and the rarity of the positive examples can lead to complicated distribution characteristics [5], which are the key factors of the imbalanced classification problem. These complicated distribution characteristics mainly include small disjuncts [6], overlapping between classes [7], rare cases and outliers [5] among the minority class space. Their simple illustrations are shown

✉ Hongjiao Guan
guan hongjiao2008@163.com

¹ School of Cyber Security, Qilu University of Technology (Shandong Academy of Sciences), Jinan 250353, China

² Shandong Computer Science Center (National Supercomputer Center in Jinan), Shandong Provincial Key Laboratory of Computer Networks, Qilu University of Technology (Shandong Academy of Sciences), Jinan 250014, China

³ School of Computer Science and Technology, Harbin Institute of Technology, Harbin 150001, China

⁴ Department of Computer Science, University of Idaho, Idaho Falls, USA

⁵ School of Computer Science, Utah State University, Logan UT, 84322, USA

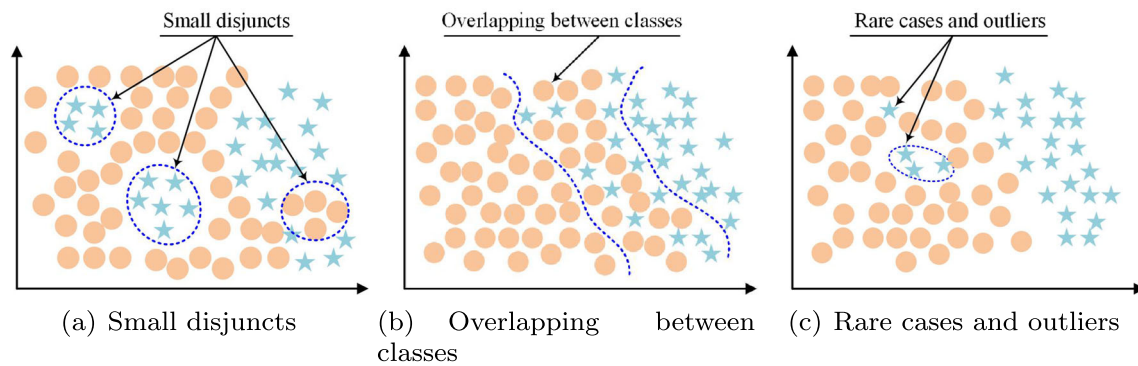


Fig. 1 Illustrations of complex distribution characteristics in imbalanced data (Circles and stars indicate the majority and minority class examples, respectively)

in Fig. 1. The study by Napierala et al. [8] indicates that the more small disjuncts and the larger the extent of overlapping between classes, the more severely the performance reduces. These complex distribution characteristics usually do not occur alone in real-world datasets [9]. This heavily hinders the correct identification of examples, especially, the minority class examples.

The classification of small-sample imbalanced datasets is even more of a challenge. When the imbalance ratio is fixed, the less the training examples, the poorer the classification performance [10]. Traditional statistical learning methods induce classification boundaries or decision rules based on the distribution of training samples. If the sample size is small, classifiers that are trained with under-represented examples cannot capture the sample distribution correctly. Moreover, the absolute rarity leads to great difficulty in classification of the minority class examples [11].

Algorithm-level and data-level methods have been proposed to solve the imbalanced classification problem. Algorithm-level methods modify traditional classification algorithms to accommodate imbalanced datasets [12, 13]. Cost-sensitive methods are to assign a large (small) misclassification cost for the minority (majority) class and train the classifier by minimizing the total misclassification cost. However, misclassification costs are difficult to obtain or estimate in practice. Empirical misclassification costs are usually set and the appropriate costs vary from datasets.

Resampling is the main data-level method, which includes undersampling, oversampling, and hybrid sampling that combines the former two. Undersampling methods remove the majority class examples randomly or strategically to remove majority class examples, such as cluster-based [14] and neighborhood-based approaches [15]. Undersampling decreases the sample size, so it is more suitable to deal with large-sample imbalanced datasets. Oversampling amplifies the minority class by copying original examples or generating synthetic examples. The popular oversampling

methods are synthetic minority oversampling technique (SMOTE) [16] and its variants such as safe-level-SMOTE [17] and local neighborhood SMOTE [18].

Hybrid sampling deletes and adds examples purposefully by considering local distribution of examples, such as SMOTE-IPF [19], SPIDER2 [8], and so on. Oversampling and hybrid sampling techniques are suitable for handling small-sample imbalanced datasets. Hybrid sampling methods that combine SMOTE and data cleaning have shown promising classification results. This kind of methods first adopt SMOTE to generate synthetic minority class examples and then use data cleaning techniques to detect and delete noisy examples. The representative methods are SMOTE-ENN [20] and SMOTE-IPF.

In small sample imbalanced datasets, the minority class examples are inclined to distribute sparsely. Although SMOTE can add minority class examples, in the areas of overlapping between classes, local densities of the two classes may still be imbalanced. Previous data cleaning methods, such as the edited nearest neighbor rule (ENN)[20], are likely to delete the minority class examples in low-density areas and compress the original positive class space. To solve such a problem, a weighted edited nearest neighbor rule (WENN) is proposed to perform data cleaning on balanced datasets after SMOTE. The basic idea of WENN is to keep positive examples in overlapping areas as much as possible by increasing the local density of the minority class. WENN adopts different scaling distances for the positive and negative examples by considering local imbalance and space sparsity. Extensive experiments over synthetic and real data validate the superiority of WENN against previous data cleaning methods. We note that a conference version of this paper appeared in [21]. This paper enriches the content about the proposed method and explains its reasonability. Furthermore, it provides a deep analysis of the distribution of preprocessed examples, which reveals the principle of performance improvement.

The rest of this paper is organized as follows. Section 2 discusses previous resampling methods related to this work. Section 3 presents the details of the proposed method. Section 4 reports and analyzes the results of extensive comparative experiments. Section 5 gives the conclusions.

2 Related works

Hybrid sampling methods based on SMOTE and data cleaning have shown effectiveness in dealing with imbalanced datasets, such as SMOTE-ENN and SMOTE-IPF. In such methods, SMOTE is first used to add minority class examples, and then data learning techniques such as ENN and IPF (iterative partitioning filter) [22] are adopted to delete noisy examples. SMOTE generates synthetic examples using linear interpolation between positive instances in the neighborhood. However, SMOTE can cause overgeneralization for the minority class [18, 19]. Figure 2 is an illustration of the overgeneralization problem of SMOTE. The new positive example (denoted by the red star) is located in the area dominated by majority class examples, which leads to overlapping between classes and blurring class boundaries.

To solve this problem, each example (called seed example) is detected whether it is a noisy one using data cleaning methods. ENN uses the local distribution to detect: if the predicted label of k -nearest neighbor (k NN) is different from the true label, the seed example is considered as noise and deleted; otherwise, the seed example is kept. IPF detects noisy examples according to the predicted results of multiple classifiers: if the majority vote of these classifiers is not the same as the true label, the seed example is deleted; otherwise, it is preserved. IPF is dependent on specific classifiers, whereas ENN relies on local distribution characteristics of the imbalanced data itself.

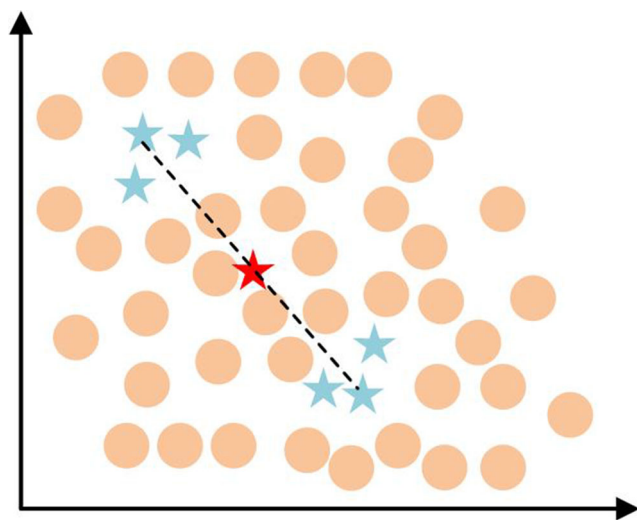


Fig. 2 Illustration of the overgeneralization problem of SMOTE

The two hybrid resampling methods (SMOTE-ENN and SMOTE-IPF) are limited when dealing with small-sample imbalanced datasets. When IPF and ENN are used to remove unsafe (or difficult to learn) minority class examples, it is uncertain whether these minority class examples are valueless. The minority class examples represent subconcepts that are underrepresented in the original imbalanced data. Thus, the minority class examples in sparse areas and overlapping areas should be treated carefully. On the other hand, for small-sample imbalanced datasets, although SMOTE has been enforced, the positive class may still have lower local distribution densities than the negative class in overlapping areas. In such areas, the minority class examples are likely to be deleted using ENN and IPF, which condenses the minority class space. Therefore, the generalization performance of the minority class will reduce.

In [15], four neighborhood-based undersampling methods are proposed. The authors claim that “the classifiers’ performance was affected more by class overlap rather than class distribution”. Therefore, their main idea is to retain all minority class examples and delete majority class examples in overlapping regions. The difference from the above hybrid sampling methods is that the methods in [15] do not add minority class examples and only perform data cleaning on the majority class. This leads to the situation that the four undersampling methods have the problem of excessive elimination of majority class examples, although different neighborhood searching criteria have been utilized. One reason is that these methods only eliminate the overlapped negative instances, which will certainly degrade the performance of the majority class. The other is that undersampling is not suitable to deal with small sample datasets, which can lead to much information loss. These issues can be seen in the following experimental results. Among the four neighborhood-based methods, NB-Comm achieves the best overall performance. Hence, NB-Comm will be compared with our proposed method in this paper.

3 Proposed method

To increase the local densities of positive examples and deal with the minority class examples in sparse areas carefully, a data cleaning method WENN is proposed. WENN improves ENN by considering *local imbalance* and *spacial sparsity*. ENN detects noisy examples according to the nearest neighbors of each seed example. Specifically, k nearest neighbors of the seed example are first obtained; then the seed example is considered as noise if the majority label of the k nearest neighbors is not consistent with the true label of the seed example.

In ENN, a heterogeneous value difference metric (HVDm) [23] is commonly used to compute the distances between examples when nominal and continuous attributes exist simultaneously. The HVDm distance is defined as follows:

$$d_{HVDm}(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{\sum_{a=1}^m d_a^2(x_{1,a}, x_{2,a})}, \quad (1)$$

where $\mathbf{x}_1$ and $\mathbf{x}_2$ denote the feature vectors of two examples, m indicates the number of attributes. $d_a(x_{1,a}, x_{2,a})$ represents the distance between x_1 and x_2 with respect to the a th attribute, which is defined as follows:

$$d_a(x_{1,a}, x_{2,a}) = \begin{cases} \sqrt{\sum_{c=1}^C \left(\frac{N_{a,x_1,c}}{N_{a,x_1}} - \frac{N_{a,x_2,c}}{N_{a,x_2}} \right)^2}, & \text{if } a \text{ is nominal;} \\ \frac{|x_1 - x_2|}{4\sigma_a}, & \text{if } a \text{ is continuous.} \end{cases} \quad (2)$$

$N_{a,x}$ is the number of training examples that have value x for attribute a . $N_{a,x,c}$ is the number of training examples that have value x for attribute a and class c . σ_a is the standard deviation of training examples for attribute a . C is the number of classes, and here, $C = 2$.

ENN determines noisy instances according to local distribution. The instances in homogeneous regions are safe and cannot be detected as noisy examples. Borderline instances or instances in heterogeneous areas are candidate noisy examples. Among these examples, positive examples are inclined to be deleted, causing the classification boundary to be biased toward the positive class space. Therefore, the recognition rate of the minority class examples reduces.

The local distribution of samples is closely related to two factors: *local imbalance* and *spatial sparsity*. Considering these two factors, the weight coefficients associated with positive and negative candidate nearest neighbors are introduced into the HVDm distance:

$$d_W^{IR,m}(\mathbf{x}_1, \mathbf{x}_2) = \begin{cases} e^{(IR^+)^m} \cdot d_{HVDm}(\mathbf{x}_1, \mathbf{x}_2), & \text{if } \mathbf{x}_2 \text{ is positive;} \\ e^{(IR^-)^m} \cdot d_{HVDm}(\mathbf{x}_1, \mathbf{x}_2), & \text{if } \mathbf{x}_2 \text{ is negative;} \end{cases} \quad (3)$$

where $\mathbf{x}_1$ indicates the seed example and $\mathbf{x}_2$ indicates a candidate nearest neighbor. IR denotes the imbalance ratio, which is defined as the number of negative examples (N^-) divided by that of positive examples (N^+); $IR = N^-/N^+$. N is the sample size of the imbalanced dataset; $N = N^+ + N^-$. IR^+ and IR^- are defined as $IR^+ = N^+/N$ and $IR^- = N^-/N$, respectively. $IR^+ + IR^- = 1$. IR can be represented as $IR = IR^-/IR^+$.

Next, the proposed weighted distance function will be analyzed. For convenience, the weight coefficients are denoted using f^+ and f^- :

$$f^+(IR, m) = e^{(IR^+)^m}, \quad (4)$$

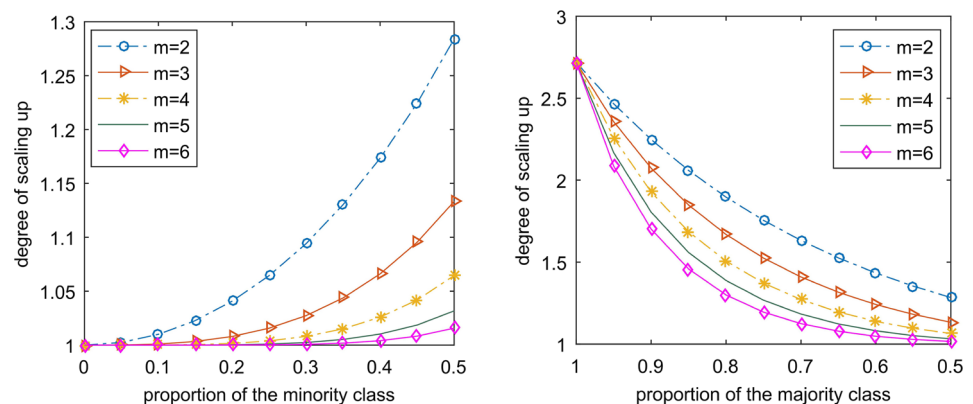
$$f^-(IR, m) = e^{(IR^-)^m}. \quad (5)$$

Figure 3 shows the change tendency of the weight coefficients f^+ and f^- with different attribute numbers and imbalance ratios. Figure 3a is the illustration of (4), where the horizontal axis represents the proportion of the minority class examples ($IR^+ \in (0, 0.5)$), and the vertical axis is the weight coefficient for positive candidate nearest neighbors (f^+). Figure 3b is the illustration of (5), where the horizontal axis represents the proportion of the majority class examples ($IR^- \in (0.5, 1)$), and the vertical axis is the weight coefficient for negative candidate nearest neighbors (f^-). Note that the x-axis is in descending order in Fig. 3b.

Local imbalance The degree of local imbalance is approximately represented by a global imbalance ratio IR . To alleviate the effect of local imbalance on determining nearest neighbors, the distances from the seed example to positive examples should be scaled smaller than to negative examples. Since $IR^+ < IR^-$ and the exponential function is increasing monotonically, when the number of attributes m is fixed, the weight coefficients satisfy the above requirement:

$$f^+(IR, m) < f^-(IR, m). \quad (6)$$

Fig. 3 Weight coefficients with respect to attribute numbers and imbalance ratios



(a) Weight of the minority class (b) Weight of the majority class

When the sample size is fixed, the larger the imbalance ratio IR , the smaller the number of minority class examples and the higher the likelihood that the minority class is decomposed into small subgroups, especially, for small datasets. Consequently, minority class examples are more dispersed. The distances between the seed example and minority class examples in the case of high IR (low IR^+ accordingly) should be scaled smaller than in the case of low IR (high IR^+ accordingly). From Fig. 3a, when m is fixed, f^+ is a monotonically increasing function of IR^+ :

$$e^{(IR_1^+)^m} < e^{(IR_2^+)^m}, \text{ if } IR_1^+ < IR_2^+. \quad (7)$$

When the sample size is fixed, the higher IR^+ , the lower IR . Therefore, f^+ satisfies the following:

$$f^+(IR_1, m) < f^+(IR_2, m), \text{ if } IR_1 > IR_2. \quad (8)$$

The case is opposite for negative candidate nearest neighbors. That is, the distances between the seed example and majority class examples in the case of high IR (high IR^- accordingly) should be scaled larger than in the case of low IR (low IR^- accordingly). From Fig. 3b, when m is fixed, f^- is a monotonically increasing function of IR^- :

$$e^{(IR_1^-)^m} < e^{(IR_2^-)^m}, \text{ if } IR_1^- < IR_2^-. \quad (9)$$

When sample size is fixed, the higher IR^- , the higher IR . Therefore, f^- satisfies

$$f^-(IR_1, m) < f^-(IR_2, m), \text{ if } IR_1 < IR_2. \quad (10)$$

Spacial sparsity When sample size is fixed, large number of feature dimensions leads to sparse example distribution. To lessen the impact of sparseness, when the number of attribute m becomes larger, the scaling coefficients should be smaller for both the majority and minority classes. From Fig. 3, when imbalance ratio is fixed, f^+ and f^- are monotonically decreasing with regard to m . It can also be analyzed according to (4) and (5). Since IR^+ and IR^- are

less than 1, when m increases, $(IR^+)^m$ and $(IR^-)^m$ reduce. Thus, f^+ and f^- decreases with the increase of m :

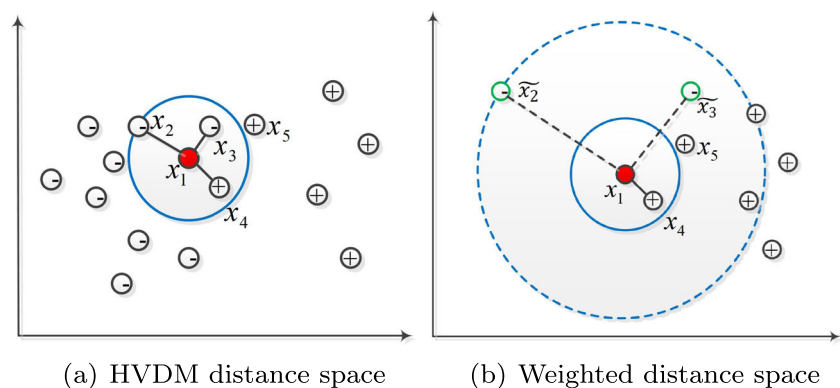
$$f^+(IR, m_1) < f^+(IR, m_2), \text{ if } m_1 > m_2; \quad (11)$$

$$f^-(IR, m_1) < f^-(IR, m_2), \text{ if } m_1 > m_2. \quad (12)$$

The principle of WENN data cleaning can be explained using Fig. 4, where x_1 indicates the seed example. Assume that three nearest neighbors are used to predict the label of the seed example. In the HVDM distance space (Fig. 4(a)), the three nearest neighbors of x_1 are x_2 , x_3 , and x_4 . x_2 and x_3 are negative examples. x_4 is a positive example. According to ENN, if x_1 is positive, it is deleted; otherwise, it is preserved. In the weighted distance space, an extreme situation is assumed: for positive candidate nearest neighbors, $IR^+ \rightarrow 0$, so $f^+ \rightarrow 1$; for negative candidate nearest neighbors, $IR^- \rightarrow 1$, so $f^- \rightarrow e$. Thus, the position of positive examples remains unchanged, whereas the weighted distances between the seed example and negative examples are about e times of the corresponding HVDM distances. x_2 and x_3 in the HVDM distance space correspond to $\tilde{x}_2$ and $\tilde{x}_3$ in the weighted distance space, respectively. In such a case, the three nearest neighbors of x_1 are $\tilde{x}_3$, x_4 , and x_5 . x_4 and x_5 are positive examples. Therefore, according to WENN, if x_1 is positive, it is preserved; otherwise, it is removed. This is the opposite of using ENN.

To summarize, the proposed weighted distance function introduces different scaling coefficients for positive and negative candidate nearest neighbors. The positive examples gather towards the seed example, and thus, the local density of the positive examples increases. By contrast, the negative examples move away from the seed example, so the local density of the negative examples decreases. Therefore, WENN is inclined to preserve the positive examples in overlapping regions and delete the negative examples. Algorithm 1 gives the pseudo-code of using SMOTE-WENN to preprocess imbalanced datasets.

Fig. 4 Illustration of WENN data cleaning



Algorithm 1 Using SMOTE-WENN to preprocess imbalanced data.

Input: Tr , the training set; p , the number of nearest neighbors in SMOTE; k , the number of nearest neighbors in data cleaning methods.

Output: New_Tr , the training set after using SMOTE-WENN.

- 1 Divide Tr into positive and negative subsets:
 $Tr = Pos \cup Neg$;
- 2 Oversample the minority class using SMOTE to balance class distribution:
 $New_Pos \leftarrow SMOTE(Pos, p)$ and
 $|New_Pos| = |Neg|$;
- 3 $New_Tr \leftarrow New_Pos \cup Neg$;
- 4 **for** $x \in New_Tr$ **do**
- 5 Compute weighted distances according to (3);
- 6 **end**
- 7 Remove noisy examples using ENN based on the weighted distances:
 $New_Tr \leftarrow ENN(New_Tr, d_w, k)$.

4 Experimental results and analysis

Experiments on synthetic and real datasets are implemented to compare the proposed SMOTE-WENN and three state-of-the-art resampling methods, including two hybrid methods (SMOTE-ENN and SMOTE-IPF) and an under-sampling method (NB-Comm).

4.1 Experimental datasets and setups

Three groups of synthetic datasets and ten real-world datasets are used in this experiment. The synthetic datasets derive from three datasets with different shapes of small disjuncts: sub-cluster, clover, and paw [8, 9]. In each group, 10 datasets are configured with two factors: IR and disturbance ratio (DR). Disturbance ratio indicates the degree of overlapping between classes or disturbing the borders of sub-regions in the minority class. Imbalanced

datasets with two IR s are considered: datasets with $IR = 5$ have 600 examples and datasets with $IR = 7$ have 800 examples. Five levels of DR are considered: 0%, 30%, 50%, 60%, and 70%, respectively. All the synthetic imbalanced datasets are two dimensions in the numerical space. Figures 5 and 6 clearly show synthetic datasets with $IR = 5$, $DR = 0\%$ and $IR = 5$, $DR = 70\%$, respectively.

Ten real-world datasets are available from the UCI repository. Their characteristics are shown in Table 1. To understand the classification difficulty of imbalanced datasets explicitly, according to the method proposed by Napierala [5], all positive (negative) examples are divided into four types: safe, borderline, rare, and outlying examples. Tables 2 and 3 list the numbers and proportions of the four type examples in the majority and minority classes, respectively. From the two tables, it can be observed that most majority class examples are safe. Several datasets have no or only about 1% rare and outlying negative examples. For minority class examples, datasets *new-thyroid* and *ionosphere* have more safe examples than unsafe examples (including borderline, rare, outlying ones). Borderline examples account for the majority in datasets *ecoli* to *transfusion*. Almost all positive examples in the last four datasets are unsafe. Especially, in dataset *balance-scale*, the vast majority of positive examples are outliers, which is difficult to classify.

Ten runs of a stratified ten-fold cross-validation are performed and the averaged results are obtained. In SMOTE, the parameter of nearest neighbors is set as 5 according to the previous recommendation [19]. The decision tree algorithm CART without pruning is used to construct the classifier. The minimum number of leaf node instances is set as 2. The classification performance is evaluated using the area under the ROC curve (AUC), sensitivity, and G-mean. AUC and G-mean are the arithmetic mean and the geometric mean of sensitivity and specificity, respectively [24]. AUC and G-mean are comprehensive and commonly used metrics in imbalanced data classification, which are not affected by class distribution [15, 25]. Sensitivity is used to evaluate the recognition rate of the minority class. When evaluating

Fig. 5 Synthetic datasets with $IR=5$ and $DR=0\%$

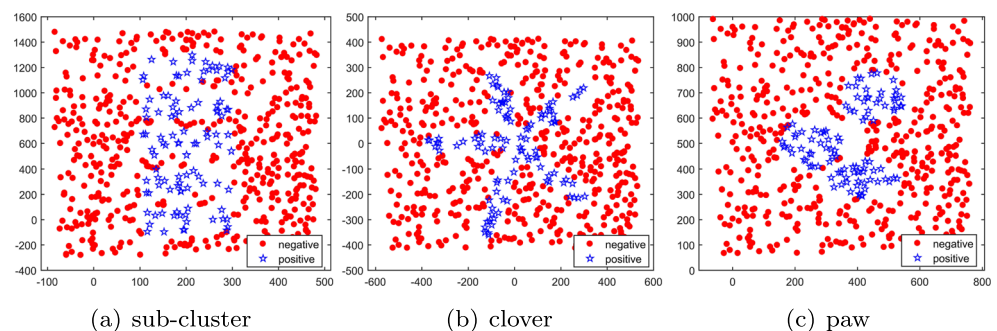


Fig. 6 Synthetic datasets with $IR=5$ and $DR=70\%$

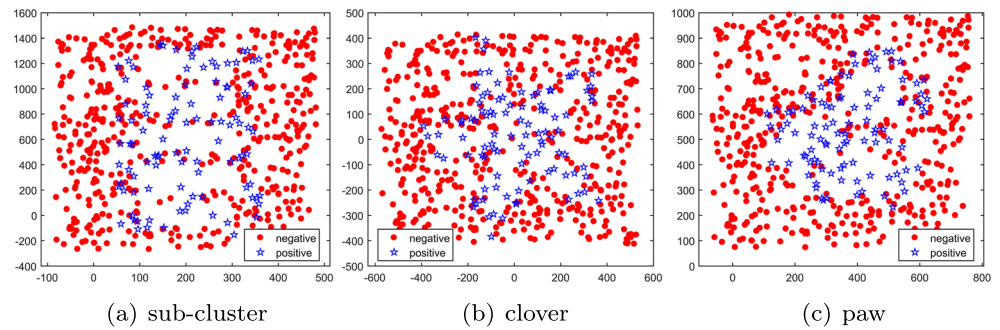


Table 1 Characteristics of 10 real datasets

Dataset	Instances	Positive	IR	Attributes
new-thyroid	215	35	5.14	5
ionosphere	351	126	1.79	34
ecoli	336	35	8.6	7
hepatitis	80	13	5.15	19
cmc	1473	333	3.42	9
transfusion	748	178	3.2	4
solar-flare	1066	43	23.79	12
yeast	1484	51	28.1	8
cleveland	297	35	7.49	13
balance-scale	625	49	11.76	4

Table 2 Numbers and ratios of four types of majority class examples

Dataset	Safe		Borderline		Rare		Outlying	
	No.	ratio(%)	No.	ratio(%)	No.	ratio(%)	No.	ratio(%)
new-thyroid	178	98.89	2	1.11	0	0.00	0	0.00
ionosphere	219	97.33	3	1.33	2	0.89	1	0.44
ecoli	280	93.02	16	5.32	3	1.00	2	0.66
hepatitis	58	86.57	9	13.43	0	0.00	0	0.00
cmc	819	71.84	283	24.82	28	2.46	10	0.88
transfusion	442	77.54	107	18.77	17	2.98	4	0.70
solar-flare	991	96.87	31	3.03	1	0.10	0	0.00
yeast	1412	98.53	18	1.26	3	0.21	0	0.00
cleveland	235	89.69	26	9.92	1	0.38	0	0.00
balance-scale	498	86.46	78	13.54	0	0.00	0	0.00

Table 3 Numbers and ratios of four types of minority class examples

Dataset	Safe		Borderline		Rare		Outlying	
	No.	ratio(%)	No.	ratio(%)	No.	ratio(%)	No.	ratio(%)
new-thyroid	24	68.57	11	31.43	0	0.00	0	0.00
ionosphere	56	44.44	33	26.19	21	16.67	16	12.70
ecoli	9	25.71	16	45.71	7	20.00	3	8.57
hepatitis	4	30.77	6	46.15	0	0.00	3	23.08
cmc	46	13.81	142	42.64	83	24.92	62	18.62
transfusion	33	18.54	64	35.96	40	22.47	41	23.03
solar-flare	1	2.33	12	27.91	13	30.23	17	39.53
yeast	2	3.92	19	37.25	10	19.61	20	39.22
cleveland	0	0.00	12	34.29	7	20.00	16	45.71
balance-scale	0	0.00	0	0.00	4	8.16	45	91.84

performance of imbalanced data, the above three metrics should be first compared. F1-score and specificity are also used mainly to compare SMOTE-WENN and NB-Comm on real datasets. All the experiments are performed using Matlab R2017a.

4.2 Results on synthetic datasets

In SMOTE-ENN and SMOTE-WENN, the nearest neighbor parameter k in data cleaning methods requires to be set. We test four values $k = 3, 5, 7, 9$ and choose the parameters of the best results: $k = 9$ in ENN and $k = 3$ in WENN.

Table 4 lists the results of AUC, sensitivity, and G-mean when using SMOTE-ENN, SMOTE-IPF, NB-Comm, and SMOTE-WENN to preprocess synthetic data sets. The best case for each dataset is highlighted in bold. Due to length limitation, SMOTE is omitted before ENN, IPF, and WENN in Table 4. The rankings of the Aligned Friedman test [26] are shown in the last row. Lower ranking means better results. It can be observed that SMOTE-WENN obtains the largest average values and the lowest rankings of AUC and G-mean. Table 5 presents the results of the Wilcoxon signed rank test associated with SMOTE-WENN (R+) and other methods (R-). The star beside the numbers indicates that there is a statistical significance between compared methods. The low p -values (< 0.05) indicate that SMOTE-WENN outperforms SMOTE-ENN, SMOTE-IPF, and NB-Comm significantly, except the sensitivity of SMOTE-WENN vs. NB-Comm.

Figures 7, 8 and 9 show the change in AUC, sensitivity, and G-mean with the increase of disturbance ratio, respectively. Overall, the results of AUC and G-mean with overlapping ($DR \neq 0\%$) are worse than those without overlapping ($DR = 0\%$). This validates the previous conclusion that class overlapping is the key factor of the classification performance [27].

It can also be observed that: (1) Compared with the two hybrid resampling methods, SMOTE-WENN has lower performance degradation. When $DR = 0\%$, SMOTE-ENN and SMOTE-WENN achieve better results of AUC and G-mean. The sensitivity of SMOTE-WENN is about five percent higher than that of SMOTE-ENN or SMOTE-IPF. This is because WENN keeps more minority class examples in overlapping areas. SMOTE-IPF obtains the smallest sensitivity, which may attribute to the unthorough data cleaning. (2) Compared with the undersampling method NB-Comm, SMOTE-WENN achieves better results of AUC and G-mean. NB-Comm obtains the highest sensitivity. However, considering the not high values of AUC and G-mean, we can infer that there exists the overgeneralization of the positive class in NB-Comm.

4.3 Results on real datasets

Parameter sensitivity is also tested on the ten real datasets. There is no uniform change tendency in evaluation metrics. This attributes to the complexity of real datasets. In this experiment, the parameter of nearest neighbors is set as $k = 3$.

Table 6 presents the results of AUC, sensitivity, G-mean, F1-score, and specificity obtained on the real datasets. The best case is highlighted in bold. Due to length limitation, SMOTE is omitted before ENN, IPF, and WENN in Table 6. The rankings of the Aligned Friedman test are shown in the last row. Lower ranking means better results. It is observed that SMOTE-WENN performs the best in terms of AUC, G-mean, and F1-score. NB-Comm achieves the best sensitivity and SMOTE-WENN ranks the second. However, NB-Comm obtains the worst specificity results. It can be concluded that the elimination of negative examples in NB-Comm is excessive. Table 7 shows the Wilcoxon signed rank test results. The star beside the numbers indicates that

Table 4 Classification results (values and rankings) of decision tree on synthetic datasets

Dataset	AUC			Sensitivity			G-mean		
	ENN	IPF	NB-Comm	WENN	ENN	IPF	NB-Comm	WENN	WENN
sub-clusIR=5,DR=0	0.9334	0.9218	0.9148	0.9350	0.9660	0.9880	1.0000	0.9640	0.9328
sub-clusIR=5,DR=30	0.8208	0.8224	0.8102	0.8172	0.8440	0.8460	0.9520	0.9000	0.8205
sub-clusIR=5,DR=50	0.7950	0.7905	0.8174	0.8072	0.8280	0.8600	0.9980	0.8940	0.7942
sub-clusIR=5,DR=60	0.8060	0.8094	0.8150	0.8164	0.8480	0.9120	1.0000	0.9100	0.8047
sub-clusIR=5,DR=70	0.7832	0.8057	0.8170	0.8170	0.8140	0.9260	0.9940	0.9340	0.7825
sub-clusIR=7,DR=0	0.9323	0.9300	0.9191	0.9380	0.9520	0.9760	1.0000	0.9700	0.9321
sub-clusIR=7,DR=30	0.8159	0.8423	0.8151	0.8241	0.8140	0.8580	0.9280	0.8940	0.8158
sub-clusIR=7,DR=50	0.7757	0.8063	0.8176	0.8023	0.7700	0.8700	0.9800	0.8840	0.7756
sub-clusIR=7,DR=60	0.7801	0.8125	0.8146	0.8064	0.7840	0.9040	0.9820	0.9040	0.7801
sub-clusIR=7,DR=70	0.7669	0.8112	0.8203	0.7967	0.7680	0.9220	0.9900	0.8920	0.7665
cloverIR=5,DR=0	0.8916	0.7619	0.8440	0.8822	0.8980	0.8120	0.9240	0.8920	0.8916
cloverIR=5,DR=30	0.8260	0.7734	0.8040	0.8366	0.8240	0.8180	0.9240	0.8920	0.8259
cloverIR=5,DR=50	0.8074	0.7613	0.8008	0.8050	0.8160	0.7800	0.8940	0.8860	0.8073
cloverIR=5,DR=60	0.7906	0.7324	0.7952	0.7994	0.8100	0.7460	0.9400	0.8960	0.7902
cloverIR=5,DR=70	0.7988	0.7593	0.7744	0.8072	0.8340	0.7660	0.8980	0.8960	0.7979
cloverIR=7,DR=0	0.9023	0.7806	0.8617	0.9021	0.8880	0.8460	0.9300	0.9100	0.9022
cloverIR=7,DR=30	0.8159	0.7593	0.8110	0.8451	0.7880	0.7880	0.9080	0.8880	0.8152
cloverIR=7,DR=50	0.7811	0.7569	0.7917	0.7994	0.7540	0.7980	0.8900	0.8540	0.7805
cloverIR=7,DR=60	0.7903	0.7428	0.7907	0.8217	0.8000	0.7740	0.9180	0.9140	0.7902
cloverIR=7,DR=70	0.7907	0.7562	0.7773	0.7896	0.7960	0.7900	0.8880	0.8760	0.7906
pawIR=5,DR=0	0.9412	0.8896	0.9162	0.9466	0.9320	0.9300	0.9520	0.9540	0.9411
pawIR=5,DR=30	0.8584	0.8275	0.8412	0.8480	0.8400	0.8180	0.9000	0.8860	0.8582
pawIR=5,DR=50	0.8478	0.8205	0.8278	0.8414	0.8580	0.8300	0.9080	0.8900	0.8476
pawIR=5,DR=60	0.8104	0.7856	0.8212	0.8218	0.8080	0.8180	0.9120	0.8700	0.8102
pawIR=5,DR=70	0.8492	0.8263	0.8354	0.8580	0.9000	0.8740	0.9420	0.9420	0.8476
pawIR=7,DR=0	0.9454	0.8917	0.9160	0.9406	0.9260	0.9280	0.9460	0.9280	0.9451
pawIR=7,DR=30	0.8493	0.8329	0.8490	0.8411	0.8240	0.8140	0.8900	0.8660	0.8489
pawIR=7,DR=50	0.8301	0.8167	0.8189	0.8431	0.8300	0.8340	0.8640	0.8960	0.8301
pawIR=7,DR=60	0.7993	0.7884	0.7996	0.8201	0.7740	0.8240	0.8580	0.8640	0.7987
pawIR=7,DR=70	0.8271	0.8006	0.8350	0.8469	0.8580	0.8620	0.9260	0.9300	0.8265
Average	0.8321	0.8072	0.8291	0.8419	0.8382	0.8504	0.9345	0.9025	0.8317
Ranking	54.25	91.53	62.7	33.52	92.5	85.38	21.20	42.92	47.57
									89.27
									73.67
									31.50

Table 5 Results of Wilcoxon signed rank test associated with SMOTE-WENN (R+) and other methods (R-) on synthetic datasets

Methods	AUC			Sensitivity			G-mean		
	R+	R-	<i>p</i> Wilcoxon	R+	R-	<i>p</i> Wilcoxon	R+	R-	<i>p</i> Wilcoxon
SM-WENN vs. SM-ENN	396.00	69.00	0.0008*	460.50	4.50	<0.0001*	358.00	107.00	0.0098*
SM-WENN vs. SM-IPF	439.00	26.00	<0.0001*	390.50	15.50	<0.0001*	436.00	29.00	<0.0001*
SM-WENN vs. NB-Comm	372.00	63.00	0.0008*	26.00	409.00	<0.0001*	449.00	16.00	<0.0001*

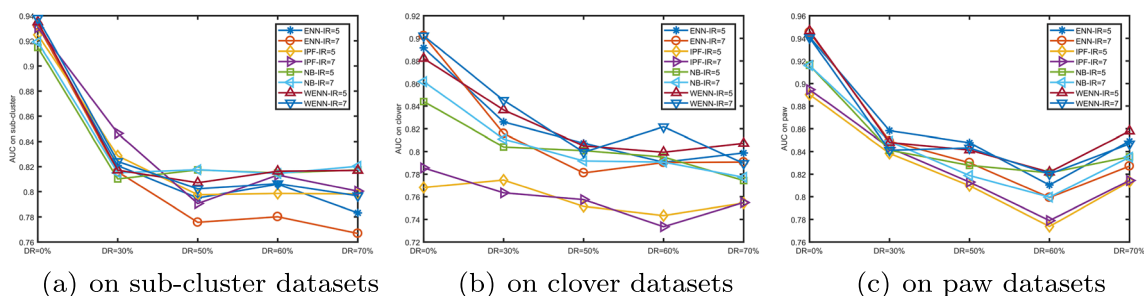
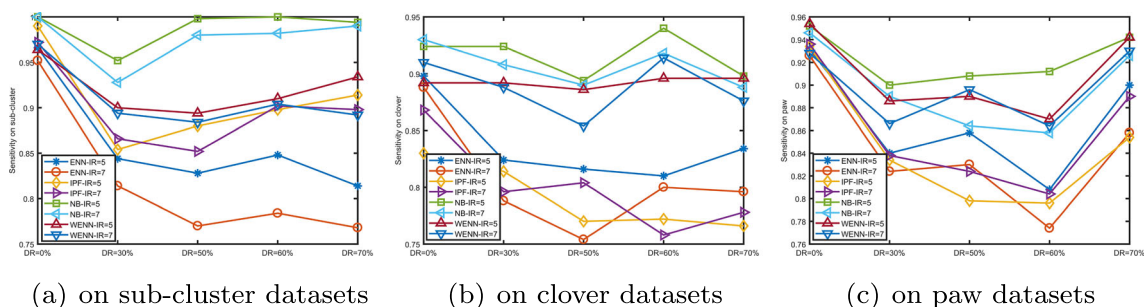
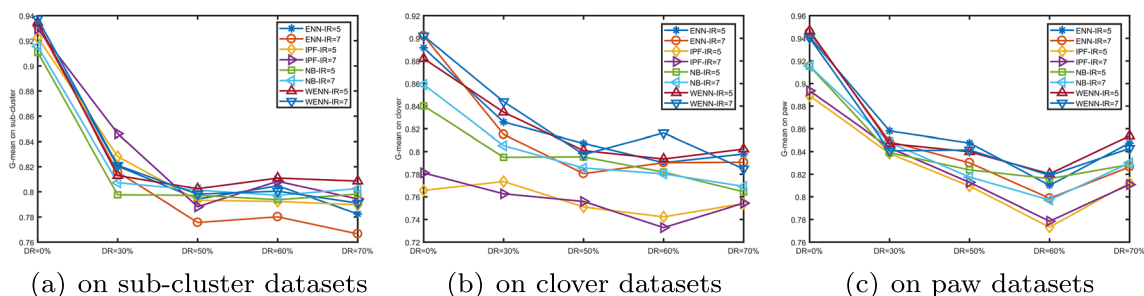
**Fig. 7** AUC results of decision tree**Fig. 8** Sensitivity results of decision tree**Fig. 9** G-mean results of decision tree

Table 6 Classification results (values and rankings) of decision tree on real datasets

Dataset	AUC				Sensitivity				G-mean			
	AUC				Sensitivity				G-mean			
	ENN	IPF	NB-Comm	WENN	ENN	IPF	NB-Comm	WENN	ENN	IPF	NB-Comm	WENN
new-thyroid	0.9182	0.9148	0.9115	0.9221	0.9086	0.9086	0.9086	0.9143	0.9181	0.9148	0.9115	0.9221
ionosphere	0.8248	0.8186	0.7382	0.8389	0.7048	0.6968	0.9492	0.7524	0.8157	0.8095	0.7071	0.8343
ecoli	0.8802	0.8765	0.8548	0.8805	0.8800	0.8686	0.8000	0.8800	0.8801	0.8764	0.8530	0.8805
hepatitis	0.5339	0.5498	0.7162	0.5488	0.7692	0.7385	0.5846	0.7692	0.4456	0.5064	0.7010	0.4978
cmc	0.6486	0.6549	0.6168	0.6609	0.6228	0.6426	0.8961	0.6396	0.6477	0.6546	0.5500	0.6603
transfusion	0.6673	0.6673	0.5953	0.6769	0.6438	0.6303	0.8573	0.6753	0.6669	0.6656	0.5344	0.6768
solar-flare	0.7874	0.7760	0.8466	0.7896	0.7070	0.6884	0.8326	0.7163	0.7827	0.7699	0.8464	0.7861
yeast	0.8123	0.8127	0.7859	0.8180	0.7882	0.7882	0.6706	0.8000	0.8119	0.8124	0.7771	0.8177
cleveland	0.6174	0.6129	0.7068	0.6229	0.5600	0.5257	0.6686	0.5543	0.6119	0.6063	0.7054	0.6188
balance-scale	0.4862	0.4742	0.5220	0.4928	0.1959	0.1755	0.2735	0.1959	0.3816	0.3674	0.4574	0.3884
Average	0.7176	0.7158	0.7294	0.7251	0.6780	0.6663	0.7441	0.6897	0.6962	0.6983	0.7043	0.7083
Ranking	21.2	22.40	20.90	17.50	22.40	24.85	14.9	19.85	21.00	22.30	20.9	17.8

Dataset	F1-score				Specificity			
	F1-score				Specificity			
	ENN	IPF	NB-Comm	WENN	ENN	IPF	NB-Comm	WENN
new-thyroid	0.8988	0.7864	0.7743	0.8866	0.9833	0.9200	0.9144	0.9722
ionosphere	0.8327	0.7410	0.6797	0.8282	0.9084	0.9396	0.5271	0.9076
ecoli	0.5329	0.5944	0.6211	0.5456	0.8751	0.8738	0.9096	0.8186
hepatitis	0.3892	0.2955	0.4958	0.4508	0.7731	0.3522	0.8478	0.8030
cmc	0.4380	0.4646	0.4304	0.4343	0.6974	0.6444	0.3375	0.6553
transfusion	0.4669	0.4883	0.4295	0.4596	0.7453	0.6902	0.3333	0.6533
solar-flare	0.2302	0.2920	0.3234	0.2968	0.9738	0.8610	0.8606	0.9589
yeast	0.2812	0.2462	0.3015	0.1860	0.9003	0.8317	0.9012	0.7213
cleveland	0.2847	0.3268	0.3744	0.3447	0.7977	0.6695	0.7450	0.7496
balance-scale	0.1141	0.0943	0.1380	0.2054	0.8094	0.6878	0.7705	0.4806
Average	0.4469	0.4330	0.4568	0.4638	0.8464	0.7470	0.7147	0.7720
Ranking	21.7	24.5	19.0	16.80	12.0	24.4	24.6	21.0

Table 7 Results of Wilcoxon signed rank test associated with SMOTE-WENN (R+) and other methods (R-) on real datasets

Methods	AUC			Sensitivity			G-mean		
	R+	R-	<i>p</i> Wilcoxon	R+	R-	<i>p</i> Wilcoxon	R+	R-	<i>p</i> Wilcoxon
SM-WENN vs. SM-ENN	55.00	0.00	0.0051*	26.50	1.50	0.0343*	55.00	0.00	0.0051*
SM-WENN vs. SM-IPF	54.00	1.00	0.0069*	54.00	1.00	0.0069*	50.00	5.00	0.0218*
SM-WENN vs. NB-Common	28.00	27.00	0.9594	18.00	37.00	0.3329	30.00	25.00	0.7989
Methods	F1-score			Specificity					
	R+	R-	<i>p</i> Wilcoxon	R+	R-	<i>p</i> Wilcoxon			
SM-WENN vs. SM-ENN	35.00	20.00	0.4446	4.00	51.00	0.0166*			
SM-WENN vs. SM-IPF	37.00	18.00	0.3329	28.00	27.00	0.9594			
SM-WENN vs. NB-Common	29.00	26.00	0.8785	36.00	19.00	0.3863			

there is a statistical significance ($p < 0.05$) between the compared methods. SMOTE-WENN outperforms SMOTE-ENN and SMOTE-IPF significantly in terms of AUC, sensitivity, and G-mean.

In order to understand the improvements achieved by resampling methods, we analyze the distributions of positive and negative examples before and after resampling. The numbers and percentages of four types (Safe, Borderline, Rare, Outlying) of positive and negative instances are calculated. The averaged results of ten times are shown in Tables 8 and 9, respectively. In the column of ‘Safe’, the cases of largest and second largest percentages are highlighted in bold and italic, respectively. In the column of ‘Total’, the cases of smallest numbers in Table 8 and largest numbers in Table 9 are highlighted in bold. In the column of ‘Method’, *None* denotes the original distribution of instances, i.e., the situation before resampling. Since NB-Comm performs data cleaning for negative examples, the numbers and percentages of only negative examples are provided in Table 8.

The resampling methods strongly change the original distributions of instances into larger (smaller) percentages of safe (unsafe) ones in each dataset. Especially, distribution change is very obvious for positive instances. For instance, for dataset *cleveland*, the percentage of safe positive instances transforms from 0% to larger than 99%.

The results in Table 8 reveal that in most cases, SMOTE-WENN retains a high percentage of safe negative examples. Compared with data cleaning method ENN, WENN removes more unsafe negative instances when dealing with balanced datasets obtained by SMOTE. For example, dataset *yeast* has only 18 negative borderline examples originally; many new negative borderline examples occur after performing SMOTE (it can be inferred from the number of negative borderline examples in the row of *SMOTE-ENN*). SMOTE-ENN reserves about 92 borderline negative instances, whereas SMOTE-WENN only has about 4 such unsafe examples. Similar cases can be observed for *ecoli*, *cmc*, *cleveland*, *balance-scale*, and so on. NB-Comm has the least total negative instances on six datasets. However, the situation of fewer negative examples means more information loss, especially, for small-sample datasets. Considering the poor specificity results of NB-Comm in Table 6, it can be concluded that NB-Comm performs excessive negative instance cleaning.

From Table 9, it is observed that SMOTE-WENN has high percentages of safe positive examples and the most total positive examples than the other two methods on most datasets. Comparing with ENN and IPF, WENN can keep more rare cases and outliers of the positive class. That is to say, SMOTE-WENN can well protect the minority class examples from being deleted blindly. Since WENN performs data cleaning in overlapping areas

Table 8 Numbers (percentages) of four types of negative examples before and after resampling

Dataset	Method	Safe	Borderline	Rare	Outlying	Total
new-thyroid	None	178(98.89)	2(1.11)	0(0)	0(0)	180
	SMOTE-ENN	172.6(98.07)	3.4(1.93)	0(0)	0(0)	176
	SMOTE-IPF	165.6(99.52)	0.8(0.48)	0(0)	0(0)	166.4
	NB-Comm	140(100)	0(0)	0(0)	0(0)	140
	SMOTE-WENN	167(100)	0(0)	0(0)	0(0)	167
ionosphere	None	219(97.33)	3(1.33)	2(0.89)	1(0.44)	225
	SMOTE-ENN	217.8(99.73)	0.2(0.09)	0.2(0.09)	0.2(0.09)	218.4
	SMOTE-IPF	206.8(99.71)	0(0)	0(0)	0.6(0.29)	207.4
	NB-Comm	46(92)	3(6)	1(2)	0(0)	50
	SMOTE-WENN	217.8(99.73)	0.2(0.09)	0.2(0.09)	0.2(0.09)	218.4
ecoli	None	280(93.02)	16(5.32)	3(1.00)	2(0.66)	301
	SMOTE-ENN	248.8(92.63)	11.8(4.39)	6.6(2.46)	1.4(0.52)	268.6
	SMOTE-IPF	255.8(95.88)	8.4(3.15)	2(0.75)	0.6(0.22)	266.8
	NB-Comm	253(98.44)	3(1.17)	1(0.39)	0(0)	257
	SMOTE-WENN	223.4(99.55)	1(0.45)	0(0)	0(0)	224.4
hepatitis	None	58(86.57)	9(13.43)	0(0)	0(0)	67
	SMOTE-ENN	52.8(92.96)	2.8(4.93)	1(1.76)	0.2(0.35)	56.8
	SMOTE-IPF	31.6(89.27)	2(5.65)	0.6(1.69)	1.2(3.39)	35.4
	NB-Comm	37(84.09)	5(11.36)	1(2.27)	1(2.27)	44
	SMOTE-WENN	52(92.86)	2.8(5.00)	1.2(2.14)	0(0)	56
cmc	None	819(71.84)	283(24.82)	28(2.46)	10(0.88)	1140
	SMOTE-ENN	577(72.56)	167(21.00)	36.2(4.55)	15(1.89)	795.2
	SMOTE-IPF	645.6(92.07)	35.6(5.08)	13.6(1.94)	6.4(0.91)	701.2
	NB-Comm	141(76.22)	28(15.14)	8(4.32)	8(4.32)	185
	SMOTE-WENN	518.2(73.73)	147.6(21.00)	24.2(3.44)	12.8(1.82)	702.8
transfusion	None	442(77.54)	107(18.77)	17(2.98)	4(0.70)	570
	SMOTE-ENN	375.2(85.78)	55.8(12.76)	3.8(0.87)	2.6(0.59)	437.4
	SMOTE-IPF	355.8(97.32)	6.2(1.70)	1.4(0.38)	2.2(0.60)	365.6
	NB-Comm	88(77.88)	12(10.62)	6(5.31)	7(6.19)	113
	SMOTE-WENN	303(84.50)	47.4(13.22)	6.2(1.73)	2(0.56)	358.6
solar-flare	None	991(96.87)	31(3.03)	1(0.10)	0(0)	1023
	SMOTE-ENN	990.6(98.16)	12.4(1.23)	3.4(0.34)	2.8(0.28)	1009.2
	SMOTE-IPF	859(99.84)	1.2(0.14)	0(0)	0.2(0.02)	860.4
	NB-Comm	825(98.92)	4(0.48)	5(0.60)	0(0)	834
	SMOTE-WENN	929.8(98.45)	10.8(1.14)	2.2(0.23)	1.6(0.17)	944.4
yeast	None	1412(98.53)	18(1.26)	3(0.21)	0(0)	1433
	SMOTE-ENN	1168.2(91.59)	91.6(7.18)	10.8(0.85)	4.8(0.38)	1275.4
	SMOTE-IPF	1108.4(93.39)	58.4(4.92)	12.2(1.03)	7.8(0.66)	1186.8
	NB-Comm	1100(99.37)	5(0.45)	2(0.18)	0(0)	1107
	SMOTE-WENN	474.6(99.00)	3.6(0.75)	1(0.21)	0.2(0.04)	479.4
cleveland	None	235(89.69)	26(9.92)	1(0.38)	0(0)	262
	SMOTE-ENN	191.6(89.70)	12.2(5.71)	6.6(3.09)	3.2(1.50)	213.6
	SMOTE-IPF	176.4(94.23)	6(3.21)	2.8(1.50)	2(1.07)	187.2
	NB-Comm	139(92.67)	10(6.67)	0(0)	1(0.67)	150
	SMOTE-WENN	179.4(95.73)	5.2(2.77)	1.4(0.75)	1.4(0.75)	187.4
balance-scale	None	498(86.46)	78(13.54)	0(0)	0(0)	576
	SMOTE-ENN	333(81.98)	71(17.48)	2.2(0.54)	0(0)	406.2
	SMOTE-IPF	364.4(91.24)	24.8(6.21)	4.6(1.15)	5.6(1.40)	399.4
	NB-Comm	280(100)	0(0)	0(0)	0(0)	280
	SMOTE-WENN	74.4(99.47)	0.4(0.53)	0(0)	0(0)	74.8

Table 9 Numbers (percentages) of four types of positive examples before and after resampling

Dataset	Method	Safe	Borderline	Rare	Outlying	Total
new-thyroid	None	24(68.57)	11(31.43)	0(0)	0(0)	35
	SMOTE-ENN	178(99.11)	1.6(0.89)	0(0)	0(0)	179.6
	SMOTE-IPF	169.4(99.76)	0.4(0.24)	0(0)	0(0)	169.8
	SMOTE-WENN	179(99.67)	0.6(0.33)	0(0)	0(0)	179.6
ionosphere	None	56(44.44)	33(26.19)	21(16.67)	16(12.70)	126
	SMOTE-ENN	152.2(79.27)	27.4(14.27)	10(5.21)	2.4(1.25)	192
	SMOTE-IPF	126.4(87.29)	11.8(8.15)	3.6(2.49)	3(2.07)	144.8
	SMOTE-WENN	152.2(79.27)	27.4(14.27)	10(5.21)	2.4(1.25)	192
ecoli	None	9(25.71)	16(45.71)	7(20.00)	3(8.57)	35
	SMOTE-ENN	296.6(99.26)	2.2(0.74)	0(0)	0(0)	298.8
	SMOTE-IPF	276.6(99.42)	1.2(0.43)	0.4(0.14)	0(0)	278.2
	SMOTE-WENN	298.6(99.20)	2.4(0.80)	0(0)	0(0)	301
hepatitis	None	4(30.77)	6(46.15)	0(0)	3(23.08)	13
	SMOTE-ENN	64.4(100)	0(0)	0(0)	0(0)	64.4
	SMOTE-IPF	55(98.92)	0.2(0.36)	0.4(0.72)	0(0)	55.6
	SMOTE-WENN	64.4(99.38)	0.2(0.31)	0.2(0.31)	0(0)	64.8
cmc	None	46(13.81)	142(42.64)	83(24.92)	62(18.62)	333
	SMOTE-ENN	878.6(89.49)	92.6(9.43)	7(0.71)	3.6(0.37)	981.8
	SMOTE-IPF	709.4(97.34)	17.2(2.36)	1.2(0.16)	1(0.14)	728.8
	SMOTE-WENN	914.8(90.18)	88(8.68)	9.2(0.91)	2.4(0.24)	1014.4
transfusion	None	33(18.54)	64(35.96)	40(22.47)	41(23.03)	178
	SMOTE-ENN	349(87.51)	37.2(9.33)	9.8(2.46)	2.8(0.70)	398.8
	SMOTE-IPF	364.8(99.35)	1.8(0.49)	0.4(0.11)	0.2(0.05)	367.2
	SMOTE-WENN	399(86.59)	40.2(8.72)	16(3.47)	5.6(1.22)	460.8
solar-flare	None	1(2.33)	12(27.91)	13(30.23)	17(39.53)	43
	SMOTE-ENN	459.6(99.91)	0.2(0.04)	0.2(0.04)	0(0)	460
	SMOTE-IPF	980.4(99.82)	0(0)	0.4(0.04)	1.4(0.14)	982.2
	SMOTE-WENN	463.8(98.81)	3.2(0.68)	0.6(0.13)	1.8(0.38)	469.4
yeast	None	2(3.92)	19(37.25)	10(19.61)	20(39.22)	51
	SMOTE-ENN	1406(99.31)	9.2(0.65)	0.2(0.01)	0.4(0.03)	1415.8
	SMOTE-IPF	1297(99.31)	8.2(0.63)	0.8(0.06)	0(0)	1306
	SMOTE-WENN	1430.6(99.86)	2(0.14)	0	0(0)	1432.6
cleveland	None	0(0)	12(34.29)	7(20.00)	16(45.71)	35
	SMOTE-ENN	254(99.76)	0.6(0.24)	0(0)	0(0)	254.6
	SMOTE-IPF	218.8(99.64)	0.4(0.18)	0(0)	0.4(0.18)	219.6
	SMOTE-WENN	256.4(99.0)	1.4(0.54)	0.2(0.08)	1(0.39)	259
balance-scale	None	0(0)	0(0)	4(8.16)	45(91.84)	49
	SMOTE-ENN	559.2(99.93)	0.4(0.07)	0(0)	0(0)	559.6
	SMOTE-IPF	260(97.23)	7.4(2.77)	0(0)	0(0)	267.4
	SMOTE-WENN	576(100)	0(0)	0(0)	0(0)	576

between classes, the sample distribution of SMOTE-WENN is more beneficial to alleviate the bias of learning algorithms towards the majority class and improve the generalization performance of the minority class.

5 Conclusions

In this paper, we propose a hybrid resampling method SMOTE-WENN to solve the problem of small-sample

imbalanced data classification. SMOTE-WENN can preserve a high percentage of safe positive and negative examples. This benefits from different distance scalings in WENN for positive and negative candidate nearest neighbors. The experimental results on synthetic and real datasets indicate that the proposed method performs significantly better than the two hybrid resampling methods (SMOTE-ENN and SMOTE-IPF). SMOTE-WENN is more suitable to handle small-sample datasets than the undersampling method NB-Comm since NB-Comm causes a lot of information loss of the negative class. The analysis of the distributions of safe, borderline, rare, and outlying examples before and after resampling, reveals the in-depth reason of the performance improvement achieved by the proposed method. The change of distribution of unsafe positive instances into more safe ones is the key factor of performance enhancement.

Acknowledgements This work was supported by the National Natural Science Foundation of China (Nos: 61872203 and 61802212), the Shandong Provincial Natural Science Foundation (No: ZR2019BF017), Major Scientific and Technological Innovation Projects of Shandong Province (Nos: 2019JZZY010127, 2019JZZY010132 and 2019JZZY010201), Jinan City “20 universities” Funding Projects Introducing Innovation Team Program (No: 2019GXRC031), and the Project of Shandong Province Higher Educational Science and Technology Program (No: J18KA331).

Compliance with Ethical Standards

Conflict of interests The authors declare that they have no conflict of interest.

References

1. Yu H, Ni J (2014) An improved ensemble learning method for classifying high-dimensional and imbalanced biomedicine data. *IEEE/ACM Trans Comput Biol Bioinform (TCBB)* 11(4):657–666
2. Yan Q, Cao Y (2020) Optimizing shapelets quality measure for imbalanced time series classification. *Appl Intell* 50(2):519–536
3. Weiss GM, Provost F (2003) Learning when training data are costly: The effect of class distribution on tree induction. *J Artif Intell Res* 19:315–354
4. Wu G, Chang EY (2005) Kba: Kernel boundary alignment considering imbalanced data distribution. *IEEE Trans Knowl Data Eng* (6):786–795
5. Napierała K, Stefanowski J (2016) Types of minority class examples and their influence on learning classifiers from imbalanced data. *J Intell Inf Syst* 46(3):563–597
6. Holte RC, Acker L, Porter BW et al (1989) Concept learning and the problem of small disjuncts. In: *Proceedings of the 11th International Joint Conference on Artificial Intelligence*, vol 89. Morgan Kaufmann Publishers, San Francisco, pp 813–818
7. Prati RC, Batista GE, Monard MC (2004) Class imbalances versus class overlapping: an analysis of a learning system behavior. In: *Mexican international conference on artificial intelligence*. Springer, Berlin, pp 312–321
8. Napierała K, Stefanowski J, Wilk S (2010) Learning from imbalanced data in presence of noisy and borderline examples. In: *International conference on rough sets and current trends in computing*. Springer, Berlin, pp 158–167
9. Stefanowski J (2013) Overlapping, rare examples and class decomposition in learning classifiers from imbalanced data. In: *Emerging paradigms in machine learning*. Springer, Berlin, pp 277–306
10. Japkowicz N, Stephen S (2002) The class imbalance problem: a systematic study. *Intell Data Anal* 6(5):429–449
11. He H, Garcia EA (2009) Learning from imbalanced data. *IEEE Trans Knowl Data Eng* 21(9):1263–1284
12. Su C, Cao J (2019) Improving lazy decision tree for imbalanced classification by using skew-insensitive criteria. *Appl Intell* 49(3):1127–1145
13. Xu Y, Wang Q, Pang X, Tian Y (2018) Maximum margin of twin spheres machine with pinball loss for imbalanced data classification. *Appl Intell* 48(1):23–34
14. Lin WC, Tsai CF, Hu YH, Jhang JS (2017) Clustering-based undersampling in class-imbalanced data. *Inf Sci* 409:17–26
15. Vuttipittayamongkol P, Elyan E (2020) Neighbourhood-based undersampling approach for handling imbalanced and overlapped data. *Inf Sci* 509:47–70
16. Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP (2002) SMOTE: synthetic minority over-sampling technique. *J Artif Intell Res* 16:321–357
17. Bunkhumpornpat C, Sinapiromsaran K, Lursinsap C (2009) Safe-level-SMOTE: Safe-level-synthetic minority over-sampling technique for handling the class imbalanced problem. In: *Pacific-asia conference on knowledge discovery and data mining*. Springer, Berlin, pp 475–482
18. Maciejewski T, Stefanowski J (2011) Local neighbourhood extension of SMOTE for mining imbalanced data. In: *2011 IEEE Symposium on Computational Intelligence and Data Mining (CIDM)*. IEEE, Washington, pp 104–111
19. Sáez JA, Luengo J, Stefanowski J, Herrera F (2015) SMOTE-IPF: Addressing the noisy and borderline examples problem in imbalanced classification by a re-sampling method with filtering. *Inf Sci* 291:184–203
20. Batista GE, Prati RC, Monard MC (2004) A study of the behavior of several methods for balancing machine learning training data. *ACM SIGKDD Explor Newsllett* 6(1):20–29
21. Guan H, Zhang Y, Xian M, Cheng HD, Tang X (2016) WENN for individualized cleaning in imbalanced data. In: *2016 23rd international conference on pattern recognition (ICPR)*. IEEE, pp 456–461
22. Khoshgoftaar TM, Rebours P (2007) Improving software quality prediction by noise filtering techniques. *J Comput Sci Technol* 22(3):387–396
23. Wilson DR, Martinez TR (1997) Improved heterogeneous distance functions. *J Artif Intell Res* 6:1–34
24. López V, Fernández A, García S, Palade V, Herrera F (2013) An insight into classification with imbalanced data: Empirical results and current trends on using data intrinsic characteristics. *Inf Sci* 250:113–141
25. Luque A, Carrasco A, Martin A, Heras ADL (2019) The impact of class imbalance in classification performance metrics based on the binary confusion matrix. *Pattern Recogn* 91:216–231
26. García S, Fernández A, Luengo J, Herrera F (2010) Advanced nonparametric tests for multiple comparisons in the design of experiments in computational intelligence and data mining: Experimental analysis of power. *Inf Sci* 180(10):2044–2064
27. Das S, Datta S, Chaudhuri BB (2018) Handling data irregularities in classification: foundations, trends, and future challenges. *Pattern Recogn* 81:674–693

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Hongjiao Guan received her PhD degree from the School of Computer Science and Technology, Institute of Technology, Harbin, China, in 2020. Now she is a lecturer at the School of Cyber Security, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China. Her research interests mainly include pattern recognition and machine learning.



Yingtao Zhang received her M.S. degree from Computer Science School of Harbin Institute of Technology, Harbin, China, in 2004, and the Ph.D. degree in Pattern Recognition and Intelligence System from Harbin Institute of Technology, Harbin, China, in 2010. Now, she is an associate professor, School of Computer Science and Technology, Harbin Institute of Technology, Harbin, China. Her research interests include pattern recognition, computer vision and medical image processing.



Min Xian received B.S. degree in Information Security from Harbin Institute of Technology, Harbin, China, in 2008, and the M.S. degree in Pattern Recognition and Intelligence System from Harbin Institute of Technology, Harbin, China, in 2011. He is now an assistant professor in the Department of Computer Science at the University of Idaho. His current research interests include biomedical data analytics, pattern recognition, machine learning, image understanding, and Neuro-Connectedness.



H. D. Cheng received Ph.D. degree in Electrical Engineering from Purdue University West Lafayette, in 1985. Now, he is a Full Professor, Department of Computer Science, and an Adjunct Full Professor, Department of Electrical Engineering, Utah State University, Logan, Utah. His research interests include image processing, pattern recognition, computer vision, artificial intelligence, medical information processing, fuzzy logic, genetic algorithms, neural networks, parallel processing, parallel algorithms, and VLSI architectures. Dr. Cheng is also an Associate Editor of Pattern Recognition, an Associate Editor of Information Sciences and Associate Editor of New Mathematics and Natural Computation.



Xianglong Tang is a professor and Ph.D. supervisor at the school of computer science and technology, Harbin Institute of Technology. His research interests include artificial intelligence and information processing.